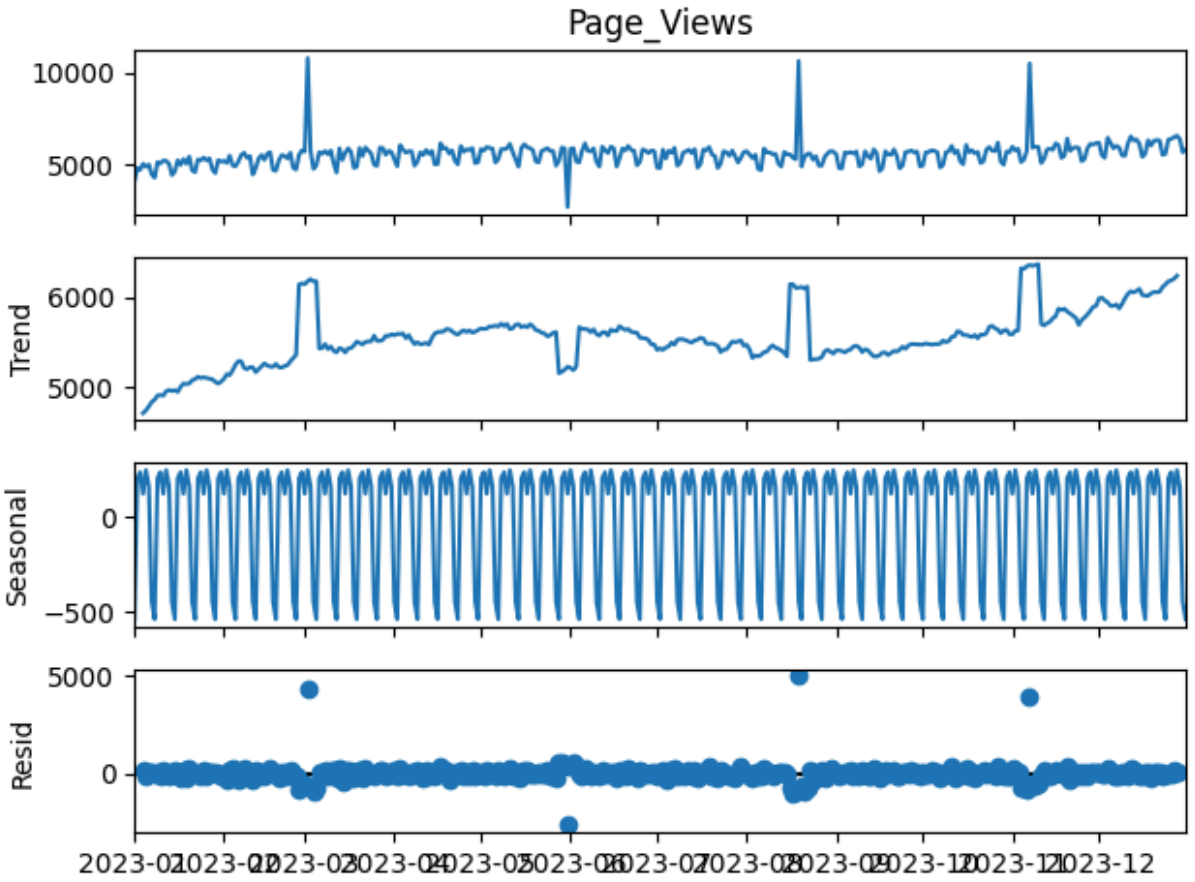
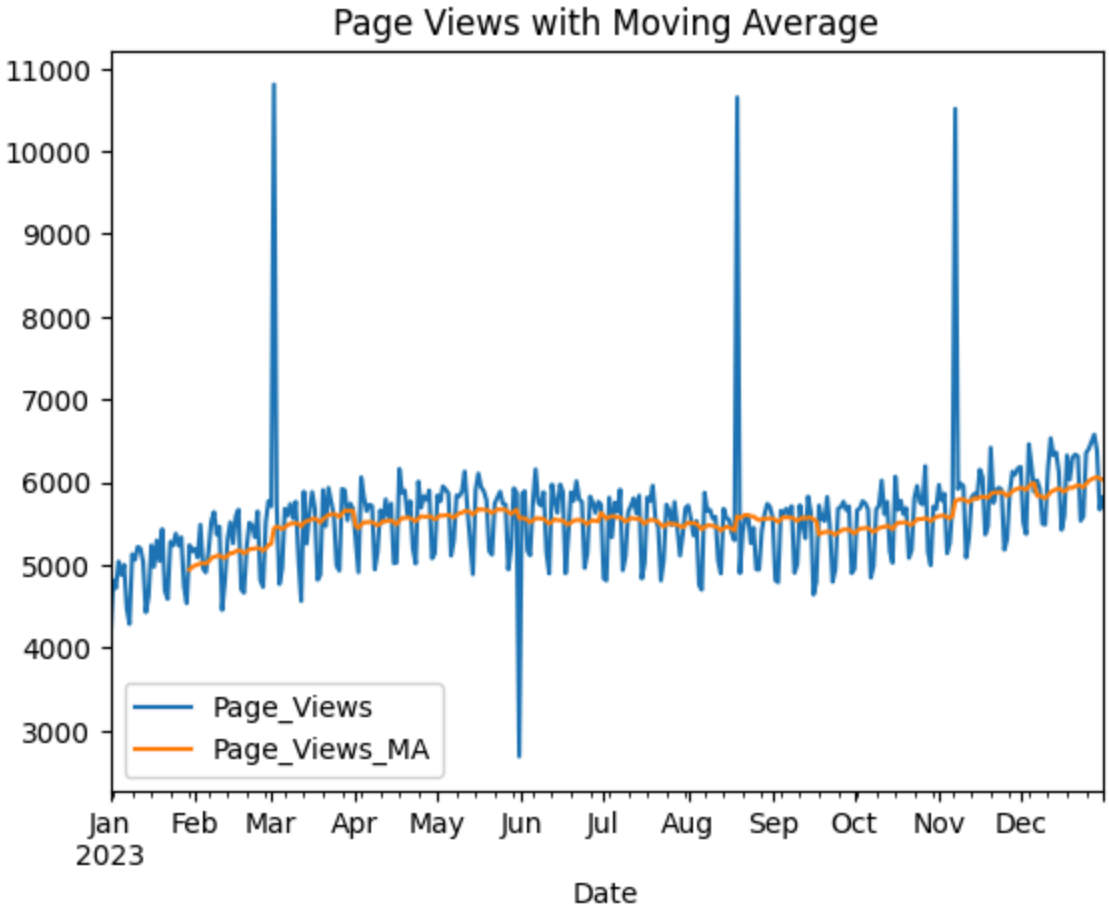
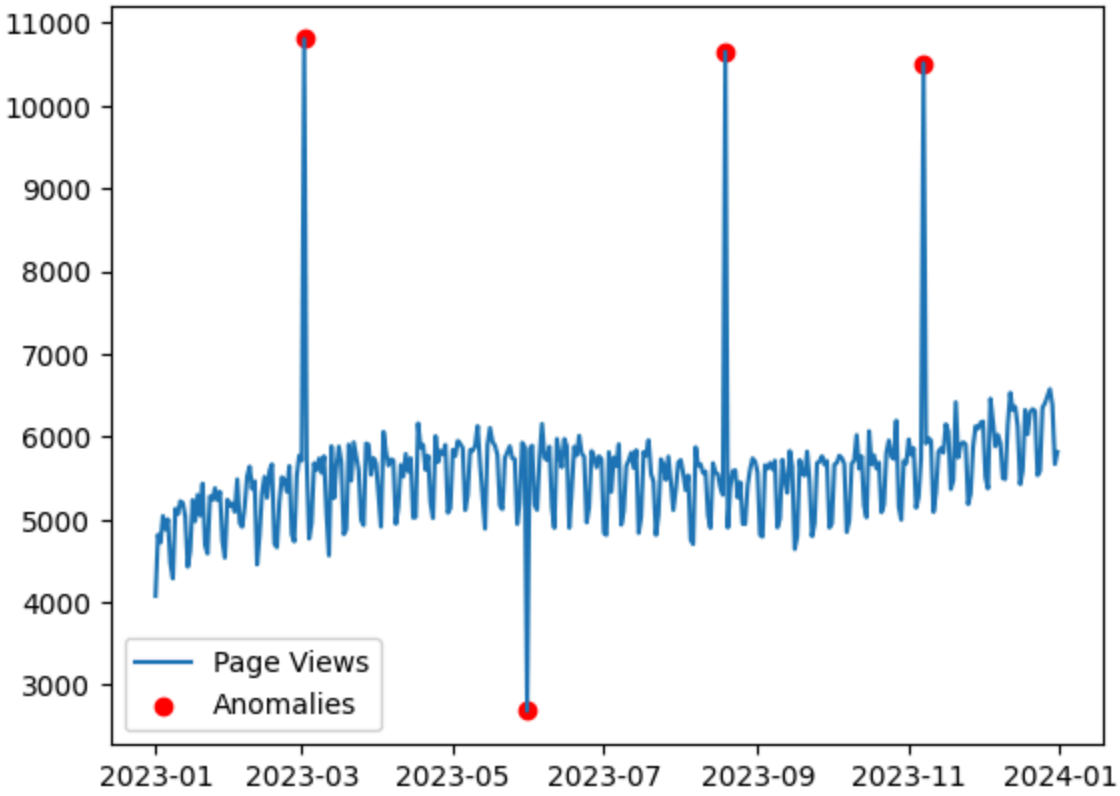
Experiment 6. Time series analysis of website traffic

In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose
data = pd.read_csv('website_traffic.csv', parse_dates=['Date'], index_col='Date')
decomposition = seasonal_decompose(data['Page_Views'], model='additive')
decomposition.plot()
plt.show()
data['Page_Views_MA'] = data['Page_Views'].rolling(window=30).mean()
data[['Page_Views', 'Page_Views_MA']].plot()
plt.title('Page Views with Moving Average')
plt.show()
data['Page_Views_Z'] = (data['Page_Views'] - data['Page_Views'].mean()) / data['Page_Views'
anomalies = data[data['Page_Views_Z'].abs() > 3]
plt.plot(data.index, data['Page_Views'], label='Page Views')
plt.scatter(anomalies.index, anomalies['Page_Views'], color='red', label='Anomalies')
plt.legend()
plt.show()
```




Page Views with Moving Average
11000
10000
9000
8000
7000
6000
5000
4000
3000
Page_Views
Page_Views_MA
Jan
2023
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Date


11000
10000
9000
8000
7000
6000
5000
4000
3000
Page Views
Anomalies
2023-01
2023-03
2023-05
2023-07
2023-09
2023-11
2024-01